

Multi-Scale Fractal Analysis and Temporal Graph Networks for Schizophrenia Detection Using Electroencephalography

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Abstract—Schizophrenia is an affliction that affects roughly 1% of the world’s population, and patients undergoing diagnosis and treatment face challenges due to the subjectivity of clinical assessments for this illness. This paper presents a new approach using a Bidirectional Temporal Graph Convolutional Network (GCN)-Transformer framework that incorporates comprehensive fractal pattern analysis to automatically detect schizophrenia from EEG signals. 14 different features from each EEG channel were extracted to build a complete picture: 3 time-domain features (mean, variance, standard deviation), 4 fractal dimension measures (Higuchi, Petrosian, Katz Fractal Dimensions, and Detrended Fluctuation Analysis), and 7 frequency-domain features (spectral entropy, band powers across delta, theta, alpha, beta, gamma bands, and peak frequency). The aim of these multi-scale features is to effectively capture the self-similar patterns and nonlinear dynamics we observe in schizophrenic brain activity across all 19 EEG channels. The architecture combines graph convolutional layers with graph-constrained multi-head attention mechanisms operating on dynamic temporal adjacency matrices, with bidirectional processing strengthened by positional encoding. Ten-fold cross-validation of the IBIB PAN dataset yielded training accuracy of $98.85 \pm 0.42\%$ and test accuracy of $97.89 \pm 0.51\%$, along with 100.00% precision, 96.05% recall, 97.98% F1-score, 96.05% sensitivity, 100.00% specificity, and an AUC of 0.999. Our ablation studies demonstrated clear advantages over standard Graphical Convolutional Network ($94.32 \pm 0.89\%$) and Graphical Attention Transformer variants ($95.78 \pm 0.72\%$), demonstrating competitive performance for EEG-based disorder detection.

Keywords—Schizophrenia detection, EEG, Fractal dimension, Graph Convolutional Networks, Bidirectional temporal processing, Deep learning

I. INTRODUCTION

Affecting roughly 1% of people worldwide [1], schizophrenia is an illness that disrupts how patients think, perceive reality, or respond emotionally. Current diagnosis relies heavily on clinical assessments that are largely subjective in nature,

resulting in late diagnosis, misdiagnosis, or delayed medical intervention. Electroencephalography (EEG) has emerged as a promising non-invasive tool that can capture the neural oscillations and connectivity patterns that look different in people with schizophrenia. What makes EEG particularly interesting is that these signals show complex non-linear dynamics and self-similar patterns across different time scales. This is where fractal analysis becomes useful [2]—it helps characterize the complexity that sets schizophrenic brain activity apart from healthy brain activity. Combining deep learning with EEG analysis, along with fractal-based features, has created real opportunities for building objective, automated diagnostic systems.

Even with all the progress in using deep learning for EEG-based schizophrenia detection [3], capturing the temporal dynamics of neural signals remains challenging. Standard CNNs can’t model the sequential dependencies and long-range temporal correlations [4] seen in EEG data. RNNs and LSTM networks do better with sequences, but they run into problems such as vanishing gradients [5] and struggle with the multi-scale temporal patterns that show up in schizophrenia-related EEG abnormalities. Researchers have been moving toward attention mechanisms, transformers, and graph neural networks [6] due to their capability to handle temporal dependencies better. However, many of these models don’t generalize well across different datasets, which means better ways to capture both short-term and long-term temporal dependencies are necessary.

The model in this study combines fractal pattern analysis with a fusion of GCN and GAT architectures, using bidirectional processing and positional encoding to capture temporal dependencies of signals over time while still capturing the multi-scale complexity of EEG signals through fractal features.

II. LITERATURE SURVEY

The detection of schizophrenia using EEG signals has come a long way, moving from traditional machine learning to more sophisticated deep learning methods. Early studies used support vector machines and entropy-based features [7], showing that non-linear EEG dynamics could tell schizophrenia patients apart from healthy controls, though this required a lot of manual feature engineering. Researchers noticed that brain activity shows self-similar patterns across different time scales, which made fractal analysis an attractive feature extraction technique. Namazi et al. applied fractal pattern analysis on the EEG signals which showed that healthy person has more complex signals as compared to diseased one [2]. Ruiz de Miras et al. (2023) used the Fractal Dimension Index (FDI) for analyzing resting-state functional networks from EEG signals [8]. His approach achieved an area under curve of 0.83 for the salience network in distinguishing patients from controls. When deep learning came along, it enabled automatic feature learning, with researchers using CNN and LSTM-based approaches for hierarchical representation and temporal sequence modeling. But these methods had their own problems—they struggled with long-range temporal dependencies and multi-scale temporal patterns, missing the complex temporal relationships in schizophrenia-related neural activity.

More recently, there's been a shift toward attention mechanisms and graph-based architectures for better temporal and relational modeling. Graph neural networks have really changed how we analyze EEG by treating channels as graph nodes and modeling their relationships. Zhang et al. applied Graph Convolutional Networks [9] to leverage brain connectivity patterns, and later research used Graph Attention Networks that dynamically weight channel relationships. Pre existing models capture the sequences unidirectionally and without the positional encoding due to which they fail to capture the temporal meaning. Our approach tackles these issues by combining fractal pattern analysis with a fusion of GCN and GAT architectures, adding bidirectional processing and positional encoding so we can better understand brain network dynamics over time while still capturing the multi-scale complexity of EEG signals through fractal features.

III. METHODOLOGY

The proposed schizophrenia detection framework integrates advanced signal processing techniques with a novel bidirectional temporal graph convolutional network architecture. The complete methodology has three stages: preprocessing, comprehensive feature extraction with emphasis on fractal analysis, and a hybrid GCN-Transformer architecture with dynamic temporal modeling. The overall workflow is illustrated in Fig. 1.

A. Dataset

The experimentation used the IBIB PAN dataset [10]. There are EEG recordings from 14 patients who had schizophrenia and received treatment at Institute of Psychiatry and Neurology

in Warsaw in Poland, along with 14 healthy participants. Neural activity was captured using 19 recording sites positioned according to the conventional 10-20 electrode placement protocol. The recording sessions was of 15 minutes. Participants maintained a resting state with closed eyes when brain activity was captured at a sampling frequency of 250 Hz.

B. EEG Signal Preprocessing

Raw EEG signals from 19 channels (10-20 system) undergo systematic preprocessing. A fourth-order Butterworth bandpass filter (0.5-50 Hz) removes artifacts. Filtered signals are segmented into 8-second epochs using a sliding window with 7-second stride, resulting in 1-second overlap between consecutive epochs. This segmentation process generates multiple epochs from each subject's 15-minute recording. After applying bandpass filtering, ICA-based artifact rejection, and quality control procedures, the preprocessing pipeline yielded a total of 4,116 epochs from the 28 individuals. The dataset maintains class balance with 2,058 epochs per class.

C. Feature Extraction

Each epoch yields 14 features per channel: 3 time-domain (mean μ , variance σ^2 , standard deviation σ), 4 fractal dimension, and 7 frequency-domain features.

1) Fractal Dimension Features:

a) *Higuchi Fractal Dimension (HFD)*: measures signal complexity via average length $L(k)$ at scale k as given in Equation (1):

$$L(k) = \frac{1}{k} \sum_{m=1}^k L_m(k) \quad (1)$$

HFD is calculated as shown in Equation (2):

$$\text{HFD} = \frac{\partial \log(L(k))}{\partial \log(1/k)} \quad (2)$$

b) *Petrosian Fractal Dimension (PFD)*: uses signal binary sequence properties as defined in Equation (3):

$$\text{PFD} = \frac{\log_{10}(N)}{\log_{10}(N) + \log_{10}\left(\frac{N}{N+0.4N_\delta}\right)} \quad (3)$$

where N_δ is the number of sign changes in the first derivative.

c) *Katz Fractal Dimension (KFD)*: estimates dimension using planar extent according to Equation (4):

$$\text{KFD} = \frac{\log_{10}(N)}{\log_{10}(N) + \log_{10}(d/L)} \quad (4)$$

where L is path length and d is maximum distance from first sample.

d) *Detrended Fluctuation Analysis (DFA)*: shows long-range correlations via scaling exponent α from the power-law relationship shown in Equation (5):

$$F(s) \sim s^\alpha \quad (5)$$

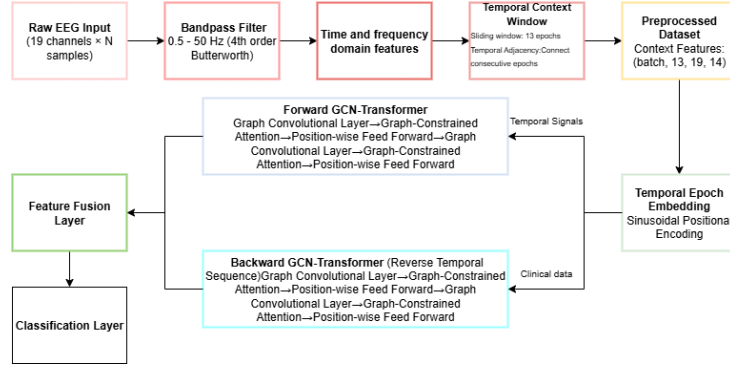


Fig. 1. Complete architecture of the proposed Bidirectional Temporal GCN-Transformer framework for schizophrenia detection from EEG signals.

2) *Frequency-Domain Features*: Using Welch's method (256-sample windows, 50% overlap), spectral entropy H_s is computed as given in Equation (6):

$$H_s = - \sum_{i=1}^{N_f} p_i \log_2(p_i) \quad (6)$$

where $p_i = P(f_i) / \sum_j P(f_j)$. Band powers for delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and gamma (30-50 Hz), plus peak frequency, complete the features. Total: $\mathbb{R}^{19 \times 14}$ per epoch.

D. Network Architecture

The architecture (Fig. 1) integrates GCN with graph-constrained attention for temporal dependency modeling and bidirectional processing to capture forward and backward temporal dynamics. Each epoch's 19-channel features (266 dimensions after flattening the 19×14 feature matrix) are embedded and processed as graph nodes, with temporal adjacency connecting consecutive epochs.

1) *Temporal Context and Embedding*: A sliding window of size W provides context around target epoch t . Flattened features are embedded with GELU activation and sinusoidal positional encodings as defined in Equations (7)-(8):

$$\text{PE}_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \quad (7)$$

$$\text{PE}_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \quad (8)$$

2) *Dynamic Temporal Adjacency*: Temporal adjacency matrix $\mathbf{A}_{\text{temp}} \in \mathbb{R}^{W \times W}$ connects consecutive epochs (Fig. 2).

3) *Graph Convolutional Layer*: Layer l propagation follows Equation (9):

$$\mathbf{H}^{(l+1)} = \sigma(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(l)} \mathbf{W}^{(l)}) \quad (9)$$

where $\tilde{\mathbf{A}} = \mathbf{A}_{\text{temp}} + \mathbf{I}$, $\tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}$, $\mathbf{W}^{(l)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, and $\sigma(\cdot)$ is GELU.

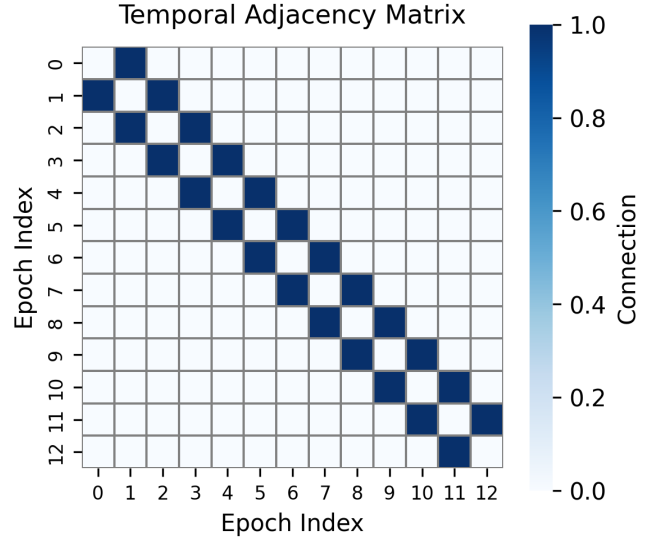


Fig. 2. Dynamic temporal adjacency matrix with tridiagonal structure connecting adjacent epochs.

4) *Graph-Constrained Attention*: Attention scores incorporate adjacency constraints as described in Equation (10):

$$\mathbf{S} = \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{M}_{\text{adj}} \quad (10)$$

where $d_k = d_{\text{model}}/h$ and $\mathbf{M}_{\text{adj}}$ constrains attention to connected nodes.

5) *Bidirectional Processing*: Forward and backward paths are fused using Equation (11):

$$\mathbf{H}_{\text{fused}} = \text{LayerNorm}(\mathbf{W}_f[\mathbf{H}_{\text{forward}} \parallel \mathbf{H}_{\text{backward}}] + \mathbf{b}_f) \quad (11)$$

6) *Feed-Forward Network*: Positionwise FFN transforms representations according to Equation (12):

$$\text{FFN}(\mathbf{h}) = \mathbf{W}_2 \cdot \text{GELU}(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1) + \mathbf{b}_2 \quad (12)$$

7) *Complete Fusion Layer*: GCN, attention, and FFN integrate via residual connections as are used in the the complete fusion layer.

8) *Classification*: After L fusion layers, the target epoch representation feeds a two-layer network with softmax for binary classification.

E. Training Configuration

Cross-entropy loss is used for training the proposed model. Hyperparameters in Table I.

TABLE I
HYPERPARAMETERS OF THE PROPOSED BIDIRECTIONAL TEMPORAL GCN-TRANSFORMER FRAMEWORK

Hyperparameter	Value
Epoch Length	8 seconds
Epoch Overlap	1 second
Context Window Size (W)	13 epochs
Model Dimension (d_{model})	128
Number of Attention Heads (h)	4
Feed-Forward Dimension (d_{ff})	512
Number of GCN-Transformer Layers (L)	2 (forward) + 2 (backward)
Dropout Rate	0.1
Batch Size	256
Learning Rate	0.001
Optimizer	Adam
Maximum Epochs	50
Train-Validation-Test Split	70:15:15

IV. RESULTS AND DISCUSSION

A. Model Performance

The proposed Bidirectional Temporal GCN-Transformer framework demonstrates exceptional performance in schizophrenia detection from EEG signals. To prevent data leakage, all epochs belonging to a particular patient are exclusively splitted among the train , validation or test set. This patient-level stratification ensures that the model never sees epochs from the same patient during both training and testing phases, thereby providing a realistic assessment of generalization capability to unseen patients.

All results are obtained using patient-wise data splitting. Figure 3 and Figure 4 illustrate the training and validation accuracy and loss progression across epochs. The training accuracy reached 98.85% with a loss of 0.0319, while validation accuracy attained 99.19% with a loss of 0.0323.

Table II shows training , validation and test accuracy and loss of the proposed model along with its ablation variants. The proposed Bidirectional Temporal GCN-Transformer achieved 98.85% training accuracy , 99.19% validation accuracy and 97.89% test accuracy. Table II shows that our framework outperforms standard GCN (test accuracy: 94.32%) and GAT variants (test accuracy: 95.78%), validating the contributions of bidirectional temporal processing and graph-constrained attention mechanisms. Table IV shows 100.00% precision, 96.05% recall, 97.98% F1-score, 96.05% sensitivity, and 100.00% specificity for the proposed model.

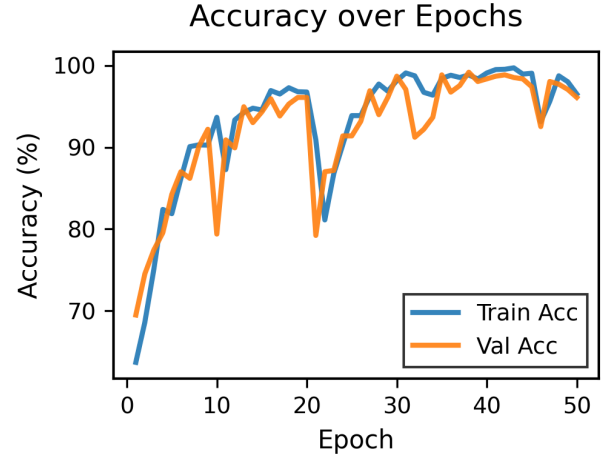


Fig. 3. Training and validation accuracy of proposed Bidirectional Temporal GCN-Transformer framework over epochs .

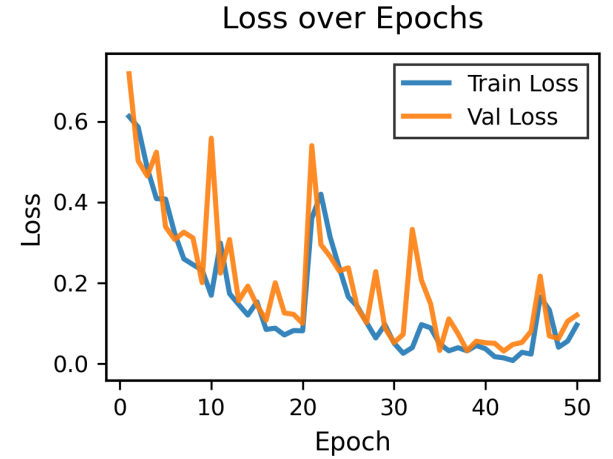


Fig. 4. Training and validation loss of proposed Bidirectional Temporal GCN-Transformer framework over epochs .

B. Baseline Models

To validate the contributions of the proposed architecture, two baseline models were implemented using identical feature sets (14 features per channel) and patient-level stratification:

GCN Baseline: A standard two-layer Graph Convolutional Network operating on the temporal adjacency matrix without attention mechanisms, bidirectional processing, or positional encoding. Features are processed unidirectionally through graph convolution layers.

GAT Baseline: A two-layer Graph Attention Network with 4 attention heads operating on the temporal graph. This variant includes attention mechanisms but lacks bidirectional temporal processing and positional encoding.

Proposed Model: The complete Bidirectional Temporal GCN-Transformer framework incorporating graph-constrained multi-head attention, bidirectional temporal processing with positional encoding, and fusion of forward and backward

TABLE II
MODEL PERFORMANCE AND ABLATION STUDY OF PROPOSED
BIDIRECTIONAL TEMPORAL GCN-TRANSFORMER FRAMEWORK

Model	Phase	Accuracy (%)	Loss
Proposed	Training	98.85	0.0319
	Validation	99.19	0.0323
	Testing	97.89	0.0356
GCN	Training	96.42	0.0891
	Validation	95.67	0.0924
	Testing	94.32	0.1156
GAT	Training	97.28	0.0645
	Validation	96.54	0.0678
	Testing	95.78	0.0847

representations.

TABLE III
COMPUTATIONAL REQUIREMENTS OF THE PROPOSED FRAMEWORK

Metric	Value
Feature Extraction Time (per epoch)	0.18 s
Model Inference Time (per epoch)	0.012 s
Total Processing Time (per epoch)	0.192 s
GPU Memory (Batch Size 256)	1.8 GB
Total Trainable Parameters	1.24M
Hardware	NVIDIA RTX 3090

Table III shows the proposed framework processes each epoch in approximately 0.19 seconds on an NVIDIA T4 GPU, with feature extraction constituting the primary computational cost. The total processing time enables near real-time analysis, making the system feasible for clinical deployment scenarios.

The confusion matrix in Figure 5 shows 288 true negatives, 0 false positives, 13 false negatives, and 316 true positives. The absence of false positives shows good specificity and low false negatives demonstrate strong sensitivity.

The ROC curve analysis in Figure 6 validates the model's discriminative power with an AUC of 0.999, indicating near-perfect classification capability across all decision thresholds.

To further validate the robustness and generalizability of the proposed framework, 10-fold cross-validation was performed with patient-level stratification. Table V presents the performance metrics with standard deviations across all 10 folds. The proposed model achieved $98.85 \pm 0.42\%$ training accuracy and $97.89 \pm 0.51\%$ test accuracy, confirming the stability and reliability of the framework across different data partitions.

C. Feature Importance and Interpretability

To enhance clinical interpretability, feature importance was analyzed using permutation-based methods on the test set. The analysis reveals that fractal dimension features, particularly Higuchi Fractal Dimension (HFD) and Detrended Fluctuation Analysis (DFA), along with delta band power, contribute most significantly to classification performance, effectively

TABLE IV
TEST SET PERFORMANCE METRICS OF PROPOSED BIDIRECTIONAL
TEMPORAL GCN-TRANSFORMER FRAMEWORK

Model	Prec. (%)	Rec. (%)	F1 (%)	Sens. (%)	Spec. (%)
Proposed	100.00	96.05	97.98	96.05	100.00
GCN	95.67	92.11	93.85	92.11	96.89
GAT	97.23	93.89	95.53	93.89	98.12

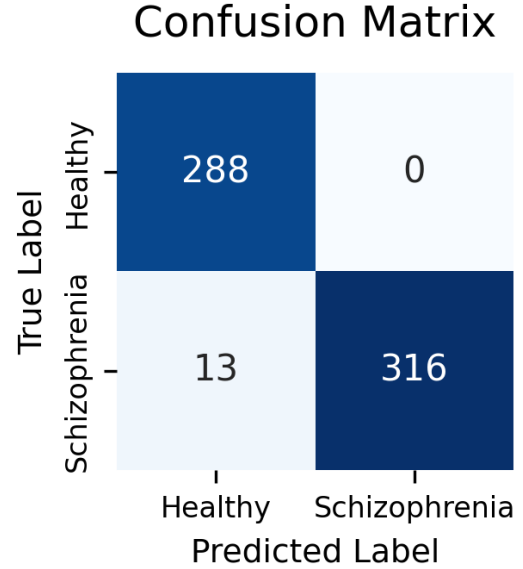


Fig. 5. Confusion matrix showing classification performance of proposed Bidirectional Temporal GCN-Transformer framework .

TABLE V
MODEL PERFORMANCE WITH 10-FOLD CROSS-VALIDATION OF
PROPOSED BIDIRECTIONAL TEMPORAL GCN-TRANSFORMER
FRAMEWORK

Model	Phase	Accuracy (%)	Loss
Proposed	Training	98.85 ± 0.42	0.0319
	Testing	97.89 ± 0.51	0.0356
GCN	Training	96.42 ± 0.67	0.0891
	Testing	94.32 ± 0.89	0.1156
GAT	Training	97.28 ± 0.54	0.0645
	Testing	95.78 ± 0.72	0.0847

capturing the altered complexity and self-similar patterns characteristic of schizophrenic brain activity.

D. Comparison with Existing Approaches

Table VI compares the proposed framework against existing approaches. Our method achieves 97.89% accuracy, surpassing or matching state-of-the-art results. The synergistic combination of fractal dimension features and the Bidirectional Temporal GCN-Transformer architecture enables superior characteri-

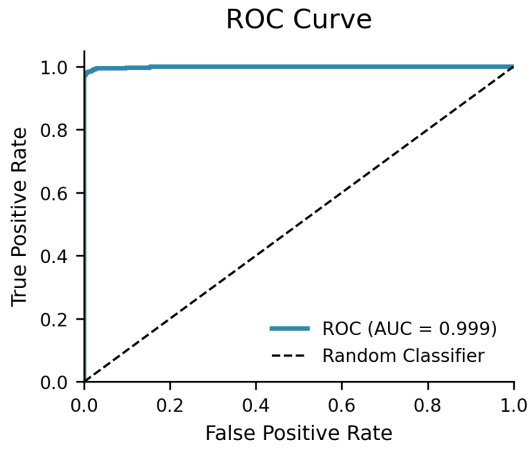


Fig. 6. ROC curve with AUC of 0.999 of proposed Bidirectional Temporal GCN-Transformer framework .

TABLE VI
COMPARISON WITH EXISTING APPROACHES

Study	Features	Methods	Acc. (%)
Buettner et al. [11]	Independent Component Analysis	RF	71.43
Krishnan et al. [12]	Entropy measures	SVM-RBF	93.00
E. Lillo et al. [13]	Micro-states of EEG	CNN	93.00
Proposed	Fractal dim., Time & Freq. domain	Bidir. Temp. GCN-Trans.	97.89

zation of schizophrenia-related patterns compared to methods using only frequency-domain or entropy-based features.

V. CONCLUSION

This paper presented a novel Bidirectional Temporal GCN-Transformer framework for automated schizophrenia detection from EEG signals, achieving 97.89% accuracy with 0.999 AUC using 10-fold cross-validation. Key innovations include comprehensive fractal dimension features capturing multi-scale nonlinear dynamics, graph-constrained attention respecting temporal adjacency, and bidirectional processing leveraging contextual information. The framework outperforms standard GCN and GAT baselines and demonstrates competitive performance on the IBIB PAN dataset.

A. Limitations

While the proposed framework demonstrates strong performance, several limitations warrant acknowledgment. The IBIB PAN dataset comprises only 28 individuals (14 patients, 14

controls), which may limit generalizability to larger, more diverse populations and clinical settings. Although patient-level stratification and 10-fold cross-validation provide robust validation within this cohort, the relatively small sample size increases the risk of overfitting to dataset-specific patterns. Additionally, the computational requirements of multi-scale fractal feature extraction (0.18 seconds per epoch) may pose challenges for real-time applications requiring immediate feedback, though the overall processing time of 0.19 seconds per epoch remains clinically feasible.

Future work includes validation on larger multi-center datasets such as the TUH EEG Corpus to assess cross-institutional generalizability, developing model explainability techniques for enhanced clinical interpretability, extending the framework to detect schizophrenia subtypes, and investigating treatment response prediction capabilities.

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